Making Devices Safe With IS 13252 STANDARD COMPLIANCE

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Electronics makes our lives easier and interesting, but it can bring in several regulations for various reasons. For example, you cannot get your electronics wet, and magnets tend to erase data from magnetic memories. Guidelines are put in place after proper testing based on product limitations. But how do you know if your gadgets are safe to use, or would not burn out in your hands even after proper safeguards?

Even the best of designs need testing

Compulsory Registration Scheme (CRS) by Bureau of Indian Standards (BIS) in collaboration with Ministry of Electronics and IT (MeitY) make sure that electronic gadgets are tested and certified before being released to the public. Among several standards under CRS, there is one governing electronic and information technology (IT) appliances such as laptops and mobile phones.

Even the best of designs have to be tested before being let out in the hands of users and maintenance people. Fires, short-circuits and radiation are some hazards involved with untested electronics. This requires testing the devices according to certain standards.

"IS 13252 talks about electrical, mechanical and fire safety of IT and electronic products from the user point of view," says Kalyan Varma, vice president, products, TUV Rheinland India. These standards, however, are only for safety requirements and do not govern the performance of devices. He adds, "It does not cover performance parameters, cyber security errors or algorithmic variations at all. It is a purely electrical safety standard."

Fig. 1: A lot of gadgets we use are covered under compliance schemes

Some guidelines for designers

IT equipment safety manual from BIS lists the following priorities that should be observed in determining what design measures to adopt:

- Where possible, specify design criteria that will eliminate, reduce or guard against hazards
- Where the above is not practicable because functioning of the equipment would be impaired, specify the use of protective means independent of equipment (such as personal protective equipment)
- Where neither of the above measures are practical, in addition to those measures, specify the provision for markings and instructions regarding residual risks

Interestingly, standards that ensure safety of the device follow minimum compliance requirements. Varma explains, "Even in the best of products, there is the possibility of residual risk." This residual risk can be reduced with improvisations in testing, but this still leaves a chance of error.

Specialised test chambers for explosions and RF. Batteries form an integral part of gadgets even though these are not covered under the specification. Hence, safety and performance of a device depend heavily on the battery as well.

Specialised chambers are required for battery testing. "We have specialised bunkers made of up to 30cm-thick walls covered with 10cm to 12cm steel for battery testing because of the possibility of explosion," says Varma. Thankfully, the ones that do explode, do not pass the test.

Electromagnetic testing of the device checks for unintentional generation, propagation and reception of electromag-